4.3 Probability

Subject content		Guidance	Curriculum reference
A	List all possible outcomes of an event and recognise equally likely outcomes	What are the possible outcomes when you roll a fair, 6-sided die?	S7.3C
B	Calculate probabilities of single and mutually exclusive events	What is the probability of rolling a 3 on a fair, 6-sided die?	S7.3D
C	Calculate the probability of an event not happening	What is the probability of NOT rolling a 3 on a fair, 6-sided die?	S7.3E
D	Estimate probability based on experimental or collected data	Given data on car colours passing the school gate, what is the probability that the next car to pass the school gate is red?	S8.3B
E	Use experimental probability to model and predict future outcomes	If 200 cars pass the school gates, how many do you expect to be red?	S8.3C

4.3 Probability *continued*

Subject content		Guidance	Curriculum reference
F	Present the possible outcomes of single events, or two successive events	(Including in lists, tables, Venn diagrams and sample space diagrams)	S9.3A
G	Calculate probabilities from possible outcomes presented in different ways	Work with tables, two-way tables, Venn diagrams and sample space diagrams	S9.3B
H	Understand when events are mutually exclusive	Are rolling a 3, and rolling an even number, mutually exclusive? You must explain your answer	S9.3C
I	Compare experimental and theoretical probabilities	Is this a fair die? (given experimental data)	S9.3D S9.3E
J	Calculate the probability of two independent events	Use P(A and B) = P(A) × P(B)	S9.3F
K	Use tree diagrams to calculate the probability of independent events		S9.3G
L	Solve problems involving probability	What is the probability that 2 identical socks picked at random from a pile are a pair?	S7.3F S8.3D S9.3H